Projected quantum kernels on up to 156 qubits for genomic classification with an independent validation cohort: preliminary results

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Working draft. This file holds the setup and the preliminary numerical results only. All results are exact noiseless simulations on the heavy-hex lattice of ibm_fez. No hardware data are included yet. The introduction, the hardware results, and the discussion are still to be written.

I. SETUP

A. Cohorts and task

The task is binary: predict the somatic *TP53* mutation status of a primary breast tumor from its gene expression. The training cohort is TCGA-BRCA [1–3], with RNA-seq expression (RSEM, transformed as $\log_2(x+1)$). The validation cohort is METABRIC [4, 5], with Illumina microarray expression. Both were obtained from cBioPortal [6, 7]. The two cohorts share 17,139 genes. A tumor is labeled positive if it carries a non-silent somatic *TP53* mutation. Table I lists the cohort sizes. The label comes from DNA sequencing and is independent of the expression data.

Two other tasks were screened with classical models and rejected (Table II). Estrogen receptor status is solved by *ESR1* alone. The PAM50 luminal A versus B label is itself computed from expression and leaves little headroom.

B. Preprocessing and gene selection

Each gene is standardized within its own cohort. This uses no labels, but it does use the unlabeled validation data. A per-sample rank normalization, which uses no cohort statistics, is reported as a control in Sec. II D. The 5,000 most variable genes of the training subset are kept. Genes are then ranked on the training subset by the ANOVA F statistic, skipping any gene whose absolute correlation with an already selected gene exceeds 0.8. The first n genes form the panel for an n-qubit circuit, so panels are nested. No label from the held-out or validation tumors enters any of these steps.

TABLE I. Cohorts. Tumors with a *TP53* call are those with both expression and sequencing data.

Cohort	Platform	Tumors	*TP53* call	Mutated (%)
TCGA-BRCA	RNA-seq (RSEM)	1,082	811	34.4
METABRIC	Illumina HT-12 microarray	1,980	1,904	34.6

TABLE II. Task screen with logistic regression. ROC AUC on held-out TCGA-BRCA and on METABRIC, 200 training tumors, mean of 3 resamples.

		Best single gene		156 genes	
Task	Top gene	TCGA	METABRIC	TCGA	METABRIC
ER status (IHC)	*ESR1*	0.957	0.957	0.951	0.941
Luminal A vs B (PAM50)	*SPAG5*	0.882	0.840	0.973	0.929
TP53 mutation	*TBC1D9*	0.864	0.791	0.925	0.874

C. Feature map and lattice

One gene is assigned to each qubit. The rotation angle of gene j is

$$x_j = \beta\,\mathrm{clip}(z_j, -3, 3), \tag{1}$$

where z_j is the standardized expression and β is the bandwidth in radians per standard deviation. The feature map repeats one block R times,

$$U(\mathbf{x}) = \Big[\prod_{(j,k)\in E} e^{-\frac{i}{2} x_j x_k Z_j Z_k} \prod_j e^{-\frac{i}{2} x_j Z_j}\; H^{\otimes n} \Big]^R, \tag{2}$$

acting on $|0\rangle^{\otimes n}$. Here E is the set of couplers of an n-qubit patch of the 156-qubit heavy-hex lattice of ibm_fez. The patch is built by removing qubits one at a time, worst first by readout error plus mean two-qubit error, never disconnecting the patch. All two-qubit gates act on native couplers, so no routing is needed. The couplers split into three layers of disjoint gates. In the CZ basis one repetition has CZ depth 6.

D. Projected quantum kernel

For each tumor we take the single-qubit expectation values $\langle P_k\rangle_{\mathbf{x}}$, $P \in \{X, Y, Z\}$, $k = 1, \ldots, n$, in the state $U(\mathbf{x})|0\rangle^{\otimes n}$. The projected quantum kernel (PQK) [8] is

$$k(\mathbf{x}, \mathbf{x}') = \exp\Big[-\gamma \sum_{k=1}^{n} \sum_{P\in\{X,Y,Z\}} \big(\langle P_k\rangle_{\mathbf{x}} - \langle P_k\rangle_{\mathbf{x}'}\big)^2\Big]. \tag{3}$$

It needs three measurement settings per tumor, independent of the number of training samples. A support vector machine is trained on this kernel.

TABLE III. Light cones of the map in Eq. (2) on all 156 qubits of ibm_fez. Cone sizes are in qubits.

R	Largest cone	Mean cone	CZ in largest cone	CZ in circuit	CZ depth
1	4	3.3	6	352	6
2	7	6.4	18	704	12
3	13	10.3	42	1,056	18
4	19	15.8	78	1,408	24
5	31	22.7	138	1,760	30
6	41	30.7	222	2,112	36

E. Baselines and protocol

The classical baselines are an RBF-kernel support vector machine (RBF-SVM), ℓ_2-regularized logistic regression (LogReg), and gradient-boosted trees (GBM), all from scikit-learn [9]. They receive the same n standardized genes as the circuit. The PQK and the RBF-SVM are tuned on the same grid of C and γ. All tuning uses 5-fold cross-validation on the training subset with ROC AUC as the score. The depth $R \in \{1, 2, 3\}$ and the bandwidth $\beta \in \{0.25, 0.5, 1\}$ of the feature map are chosen the same way.

Each resample draws 200 training tumors from TCGA-BRCA, 200 further TCGA-BRCA tumors as the held-out set, and 400 METABRIC tumors as the independent set, all stratified by class. Results are means over 5 resamples. The resamples share the same two cohorts, so the quoted standard errors are optimistic.

F. Exact simulation

The expectation value $\langle P_k \rangle$ depends only on the gates in the backward light cone of qubit k. The ZZ rotations of one repetition commute, so the cone grows by one lattice step per repetition. Table III gives the cone sizes on the full 156-qubit lattice. For $R \leq 4$ every single-qubit reduced density matrix is a statevector calculation on at most 19 qubits. All results below use this method and are exact up to floating-point error.

The simulator was checked in two ways. On 16-qubit patches it agrees with a full statevector simulation in Qiskit [10] to within 2×10^{-13}. On the full 156-qubit circuit at $R = 2$, a matrix-product-state simulation converges to the same values: the largest deviation is 3.1×10^{-2}, 4.5×10^{-4}, and 1.9×10^{-5} at bond dimension 8, 32, and 64.

II. PRELIMINARY RESULTS

A. Scaling from 20 to 156 qubits

Figure 1 and Table IV compare the PQK with the classical baselines as the panel grows from 20 to 156 genes. The PQK tracks the RBF-SVM and logistic regression on both cohorts. The paired differences in Table V stay within 0.017 in AUC and change sign from size to size; their standard errors range from 0.001 to 0.021. Logistic regression on all 5,000 prefiltered genes reaches 0.918 on held-out TCGA-BRCA and 0.890 on METABRIC.

B. Depth and bandwidth

Table VI and Fig. 2 show the effect of the feature-map hyperparameters, averaged over sizes. Cross-validation selects $\beta = 0.25$ in 28 of 35 cases and $\beta = 0.5$ in 7. The largest bandwidth, $\beta = 1$, is selected 0 times. At $\beta = 0.25$ the mean Bloch-vector length of the single-qubit states is 0.98 to 0.99, so the qubits are close to a product state. At $\beta = 1$ it falls to 0.69–0.75 and the METABRIC AUC is lower by 0.03 to 0.06. The geometric difference [8] between the PQK and the tuned RBF kernel, computed with regularization $1/C$, averages 1.1 to 1.9 across configurations and does not exceed 8.2 in any single run. The scale for a possible prediction advantage is $\sqrt{N} \approx 14$ at $N = 200$ training samples.

C. Concentration

At fixed depth and bandwidth, the mean Bloch-vector length varies by at most 0.06 and the spread of the features by at most 0.02 as the number of qubits goes from 20 to 156. This follows from locality: each feature depends only on its light cone, whose size is bounded at fixed R (Table III). The relative spread of pairwise distances decreases slowly with n for the PQK features and for the raw classical features alike.

D. Entanglement ablation

Table VII adds two controls. The first removes every two-qubit gate from the feature map, which leaves a product state. The second gives the RBF-SVM inputs clipped at three standard deviations, as the circuit receives them. Under cohort-wise standardization all five models agree within their standard errors. Under per-sample rank normalization the kernel models are less stable. In one of the five resamples the plain RBF-SVM loses more than 0.1 in METABRIC AUC at 100 and at 156 qubits, and the PQK does so at 100 qubits. The two controls and logistic regression are affected less. In every column the PQK and its entangler-free version agree within one standard error.

E. Classical simulability

Every result in this draft was obtained classically (Sec. I F). At the depths that cross-validation selects, the largest light cone holds at most 13 qubits. Cones beyond direct statevector simulation require $R \geq 5$ (Table III).



FIG. 1. ROC AUC against the number of qubits, which equals the number of genes in the panel. Left: held-out TCGA-BRCA tumors. Right: the independent METABRIC cohort. The PQK is computed by exact noiseless simulation, with depth and bandwidth chosen by cross-validation on the training set. Points are means over 5 resamples and bars are one standard error. Points are offset horizontally for clarity.

TABLE IV. ROC AUC, mean over 5 resamples. The number in parentheses is the standard error in units of the last digit.

	Held-out TCGA-BRCA				METABRIC			
Qubits	PQK	RBF-SVM	LogReg	GBM	PQK	RBF-SVM	LogReg	GBM
20	0.909(8)	0.904(9)	0.911(8)	0.887(10)	0.884(10)	0.873(17)	0.880(10)	0.861(10)
40	0.907(14)	0.915(9)	0.914(9)	0.889(8)	0.881(9)	0.888(5)	0.887(5)	0.872(8)
60	0.910(10)	0.917(9)	0.911(9)	0.885(20)	0.869(28)	0.886(8)	0.878(12)	0.858(11)
80	0.915(10)	0.912(11)	0.916(11)	0.894(11)	0.875(17)	0.869(17)	0.870(16)	0.858(10)
100	0.913(10)	0.919(9)	0.917(10)	0.894(12)	0.882(8)	0.883(4)	0.875(10)	0.854(11)
127	0.914(12)	0.917(9)	0.912(12)	0.900(14)	0.866(17)	0.876(10)	0.863(24)	0.865(8)
156	0.917(10)	0.917(10)	0.918(12)	0.900(13)	0.888(8)	0.881(8)	0.879(6)	0.856(10)

TABLE V. Difference in ROC AUC between the PQK and a classical model, paired by resample. Mean ± standard error.

	PQK − RBF-SVM		PQK − LogReg	
Qubits	TCGA	METABRIC	TCGA	METABRIC
20	$+0.005 \pm 0.008$	$+0.011 \pm 0.008$	-0.002 ± 0.002	$+0.004 \pm 0.002$
40	-0.008 ± 0.006	-0.007 ± 0.005	-0.007 ± 0.007	-0.007 ± 0.005
60	-0.007 ± 0.007	-0.017 ± 0.021	-0.001 ± 0.006	-0.009 ± 0.018
80	$+0.003 \pm 0.002$	$+0.006 \pm 0.002$	-0.001 ± 0.002	$+0.005 \pm 0.006$
100	-0.006 ± 0.004	-0.001 ± 0.005	-0.004 ± 0.003	$+0.007 \pm 0.006$
127	-0.002 ± 0.004	-0.010 ± 0.007	$+0.002 \pm 0.002$	$+0.004 \pm 0.009$
156	-0.000 ± 0.001	$+0.007 \pm 0.002$	-0.000 ± 0.004	$+0.009 \pm 0.006$

F. Shot budget and hardware cost

No hardware runs are included in this draft. This subsection records the planning numbers.

Table VIII shows a first-order noise proxy at the selected feature maps. Each expectation value is multiplied by $(1-\epsilon)^{m_k}$, where m_k is the number of CZ gates in the light cone of qubit k and $\epsilon = 0.0039$ is the median CZ error of the calibration snapshot, and is then sampled with a finite number of shots per basis. The proxy ignores crosstalk, leakage, and readout bias.

The circuits are short. At $R = 2$ on 156 qubits the transpiled circuit lasts 1.5 μs, against a default repetition delay of 250 μs. The cost is therefore set by the number of samples and shots, not by the number of qubits or the depth. With three measurement bases, 600 tumors at 1024 shots take an estimated 12 min of processor time per configuration, and 47 min at 4096 shots. These estimates include a factor 1.5 for overhead, which is a guess to be replaced by the billed time of a pilot run. The planned runs use readout mitigation only [11], with the feature map transpiled once and all tumors supplied as parameter values.

TABLE VI. PQK by depth R and bandwidth β, mean over resamples and sizes. $\overline{|\mathbf{r}|}$ is the mean Bloch-vector length, "Spread" is the mean standard deviation of an expectation value across training samples, g is the geometric difference to the tuned RBF kernel with regularization $1/C$, and "Selected" counts how often cross-validation chose the configuration.

R	β	TCGA	METABRIC	$\overline{\lvert\mathbf{r}\rvert}$	Spread	g	Selected
1	0.25	0.916	0.882	0.994	0.095	1.12	17
1	0.5	0.912	0.878	0.927	0.191	1.21	2
1	1	0.885	0.852	0.692	0.258	1.81	0
2	0.25	0.915	0.880	0.994	0.113	1.24	6
2	0.5	0.908	0.880	0.936	0.228	1.32	1
2	1	0.860	0.819	0.752	0.328	1.93	0
3	0.25	0.915	0.881	0.981	0.207	1.23	5
3	0.5	0.902	0.870	0.883	0.320	1.59	4
3	1	0.874	0.823	0.698	0.351	1.28	0

TABLE VII. ROC AUC on METABRIC under two normalizations. Mean over 5 resamples, standard error in parentheses.

Model	$n = 20$	$n = 100$	$n = 156$
Cohort-wise z-score			
PQK	0.884(10)	0.880(8)	0.886(9)
PQK, entanglers removed	0.882(9)	0.880(8)	0.881(10)
RBF-SVM	0.873(17)	0.883(4)	0.881(8)
RBF-SVM, clipped inputs	0.881(10)	0.884(4)	0.883(8)
Logistic regression	0.880(10)	0.875(10)	0.879(6)
Per-sample rank			
PQK	0.877(5)	0.848(36)	0.876(8)
PQK, entanglers removed	0.877(5)	0.868(20)	0.871(14)
RBF-SVM	0.876(6)	0.850(39)	0.860(24)
RBF-SVM, clipped inputs	0.876(5)	0.870(17)	0.875(12)
Logistic regression	0.873(6)	0.877(7)	0.874(11)

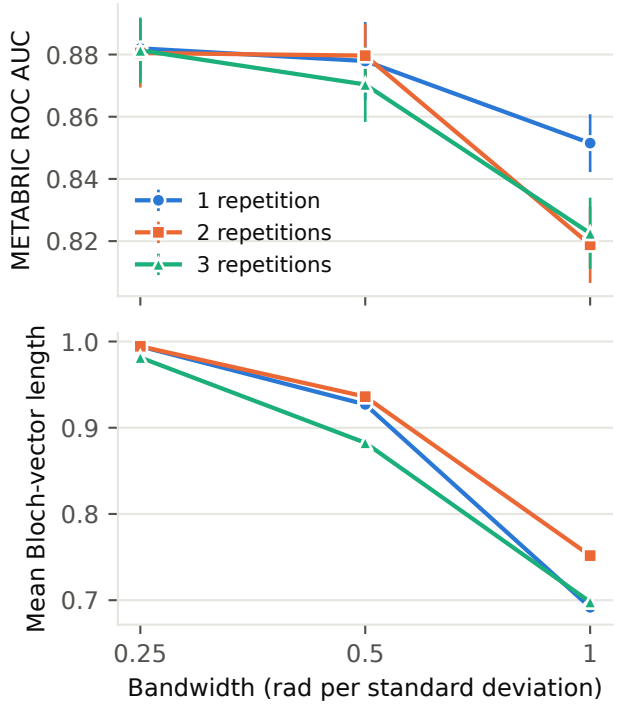


FIG. 2. METABRIC ROC AUC (top) and mean Bloch-vector length (bottom) against the bandwidth β, averaged over sizes. Bars are one standard error over resamples.

TABLE VIII. Noise proxy: ROC AUC on METABRIC at the selected feature maps, against the number of shots per measurement basis. Mean over 5 resamples.

Qubits	Exact	4096 shots	1024 shots	256 shots
20	0.884	0.884	0.875	0.882
40	0.881	0.889	0.885	0.880
60	0.869	0.865	0.866	0.859
80	0.875	0.877	0.872	0.890
100	0.882	0.878	0.879	0.882
127	0.866	0.865	0.876	0.872
156	0.888	0.883	0.887	0.888

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